

Wazuh SIEM Home Lab

Part 3: Attack Simulation with Metasploit & Nmap

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Cybersecurity Portfolio Project

April 2026

Tools Used: Wazuh SIEM | Metasploit Framework v6.4 | Nmap 7.94 | Python 3 | Ubuntu 24.04 | Windows

1. Project Overview

This document details the third phase of a Wazuh SIEM home lab portfolio project. Having deployed a full Wazuh all-in-one stack in Part 1 and built a Python automated alert reporter in part 2, this phase focuses on generating real, detectable intrusion events using industry-standard offensive security tools, such as Metasploit Framework and Nmap.

The goal is not to compromise a system, but to simulate realistic attacker behavior and observe how a SIEM detects, categorizes, and alerts on that behavior. Every attack scenario in this lab has a corresponding Wazuh alert, demonstrating the complete attack-detect-report cycle.

1.1 Learning Objectives

- Gain hands-on experience with Metasploit Framework in a controlled environment
- Understand how common attack patterns (reconnaissance, brute force) generate Windows Security Events
- Observe how Wazuh correlates raw events into meaningful alerts using its rule engine
- Demonstrate the MITRE ATT&CK framework in practice through real detection output
- Extend the Python alert reporter built in Part 1 to capture attack-generated events

1.2 Lab Architecture

Due to hardware constraints, the lab uses the following unconventional but functional architecture:

Component	Details
Host Machine	HP Laptop, Windows 11, 16GB RAM, AMD Ryzen AI 5
Hypervisor	Oracle VirtualBox
Wazuh VM	Ubuntu 24.04 LTS, 6GB RAM, IP 10.0.2.15 (NAT)
Network Type	VirtualBox NAT — VM reaches host via gateway 10.0.2.2
Monitored Agent	Windows 11 host (Sean_Murphy-PC), IP 192.168.68.85
Attacker Role	Wazuh VM (Ubuntu) — Metasploit + Nmap installed here
Attack Target	Windows 11 host — Wazuh agent enrolled and reporting
Wazuh Stack	Manager + OpenSearch Indexer + Dashboard (all-in-one)

Note on Architecture: Running the attacker tools on the same VM as the SIEM is unconventional but valid for a constrained home lab. The important outcome — simulated attacks generating detectable alerts on a monitored Windows endpoint — is fully achieved. The attack/detect story is intact.

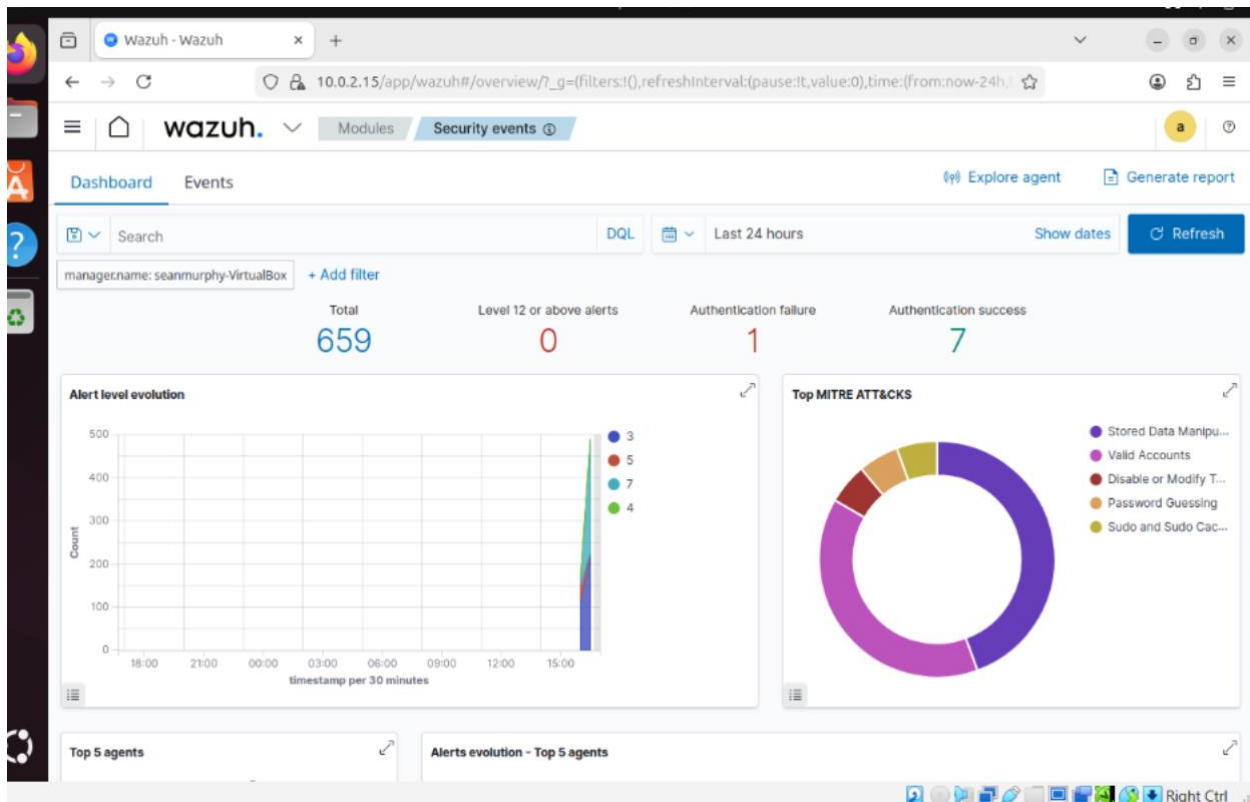


Figure: Wazuh all-in-one stack confirmed operational on Ubuntu 24.04 VM prior to attack simulation. Manager, Indexer, and Dashboard all running.

2. Environment Setup

2.1 Wazuh Stack Verification

Before any attack simulation, the Wazuh stack was verified as fully operational. All core daemons were confirmed running using the Wazuh control script:

```
sudo /var/ossec/bin/wazuh-control status
```

Key daemons confirmed running: wazuh-db, wazuh-analysisd, wazuh-remoted, wazuh-logcollector, wazuh-modulesd, wazuh-authd, wazuh-execd. The Wazuh dashboard was accessible at <https://10.0.2.15> and the Windows agent (Sean_Murphy-PC) was confirmed active.

2.2 Installing Nmap

Nmap was installed on the Ubuntu VM to serve as the network reconnaissance tool, simulating the first phase of an attacker's methodology:

```
sudo apt update && sudo apt install -y nmap  
nmap --version
```

Nmap version 7.94SVN was installed and confirmed. Nmap is used in the reconnaissance and vulnerability scanning scenarios.

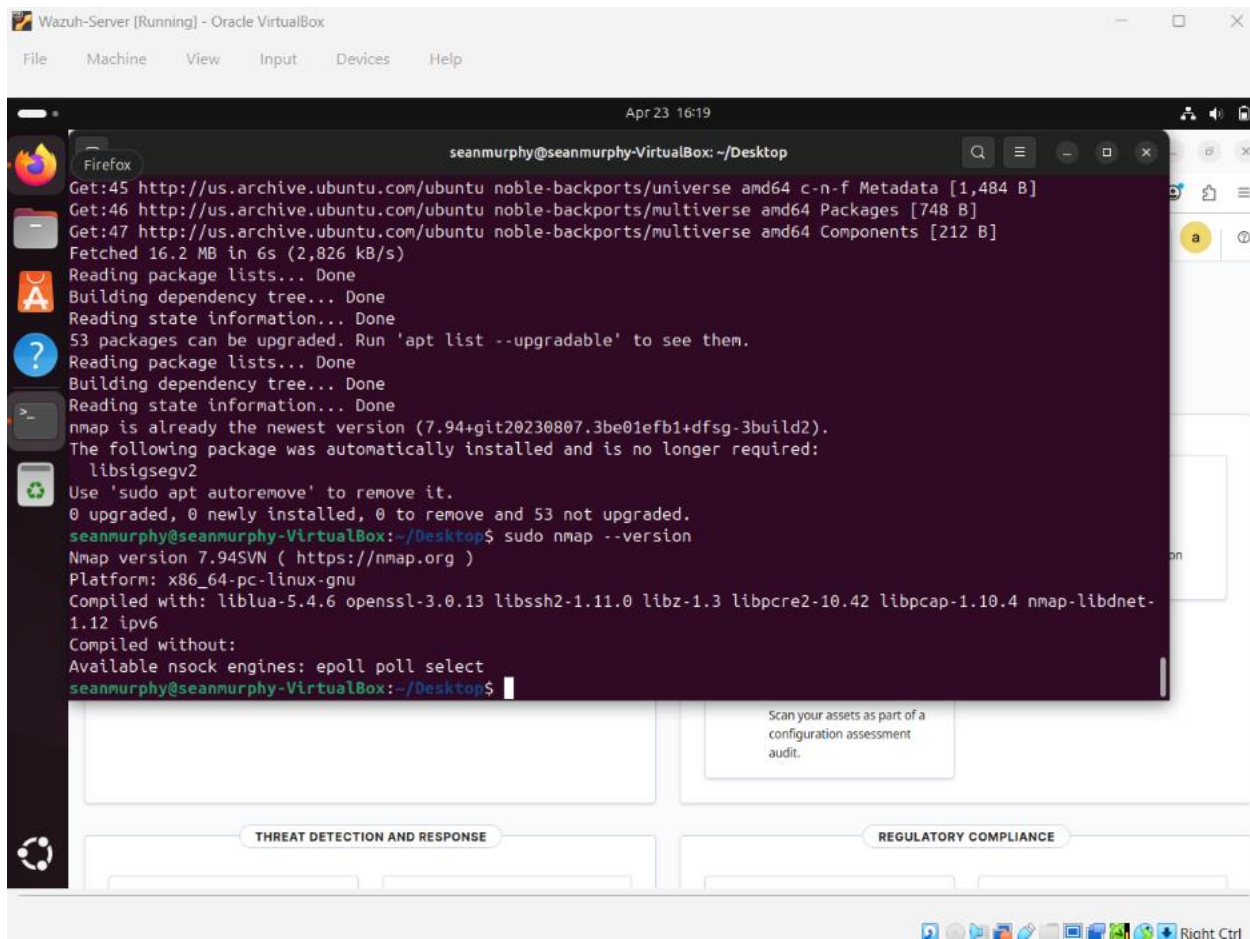


Figure: Nmap 7.94SVN installed on Wazuh VM (Ubuntu 24.04). Used to simulate network reconnaissance and vulnerability scanning attack phases.

2.3 Installing Metasploit Framework

Metasploit was installed using the official Rapid7 omnibus installer, which bundles a self-contained Ruby runtime and PostgreSQL backend. This approach avoids dependency conflicts with the Wazuh stack running on the same VM.

```
curl https://raw.githubusercontent.com/rapid7/metasploit-omnibus/master/config/templates/metasploit-framework-wrappers/msfupdate.erb > msfinstall
chmod 755 msfinstall
sudo ./msfinstall
```

After installation, the Metasploit database was initialized using the bundled PostgreSQL instance:

```
msfdb init
```

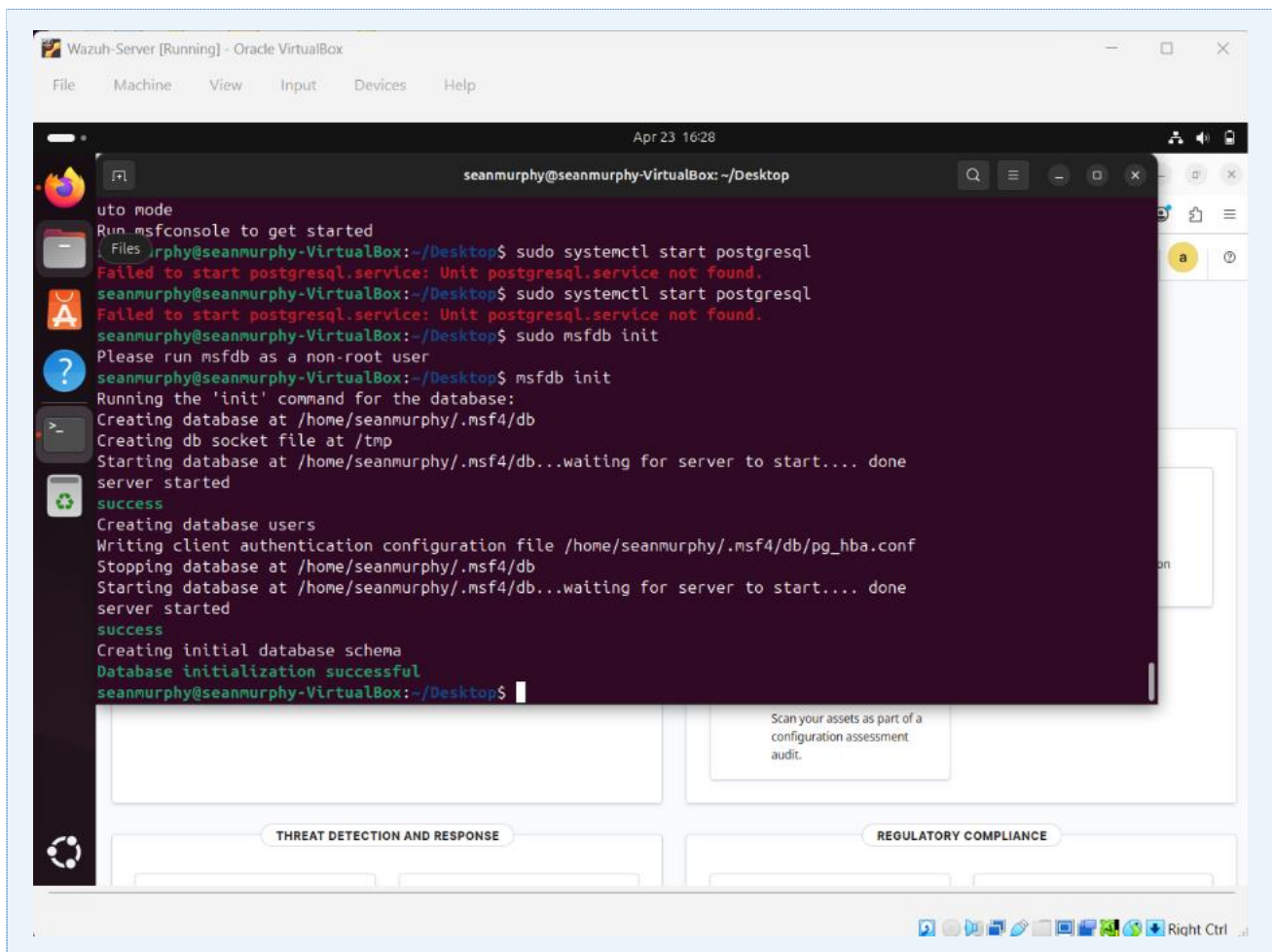


Figure: Metasploit Framework database initialized successfully. The bundled PostgreSQL backend enables session tracking, host data storage, and loot management across engagements.

Metasploit Framework v6.4.126 was launched and the database connection was verified:

```
msfconsole  
  
msf6 > db_status  
[*] Connected to msf. Connection type: postgresql.
```

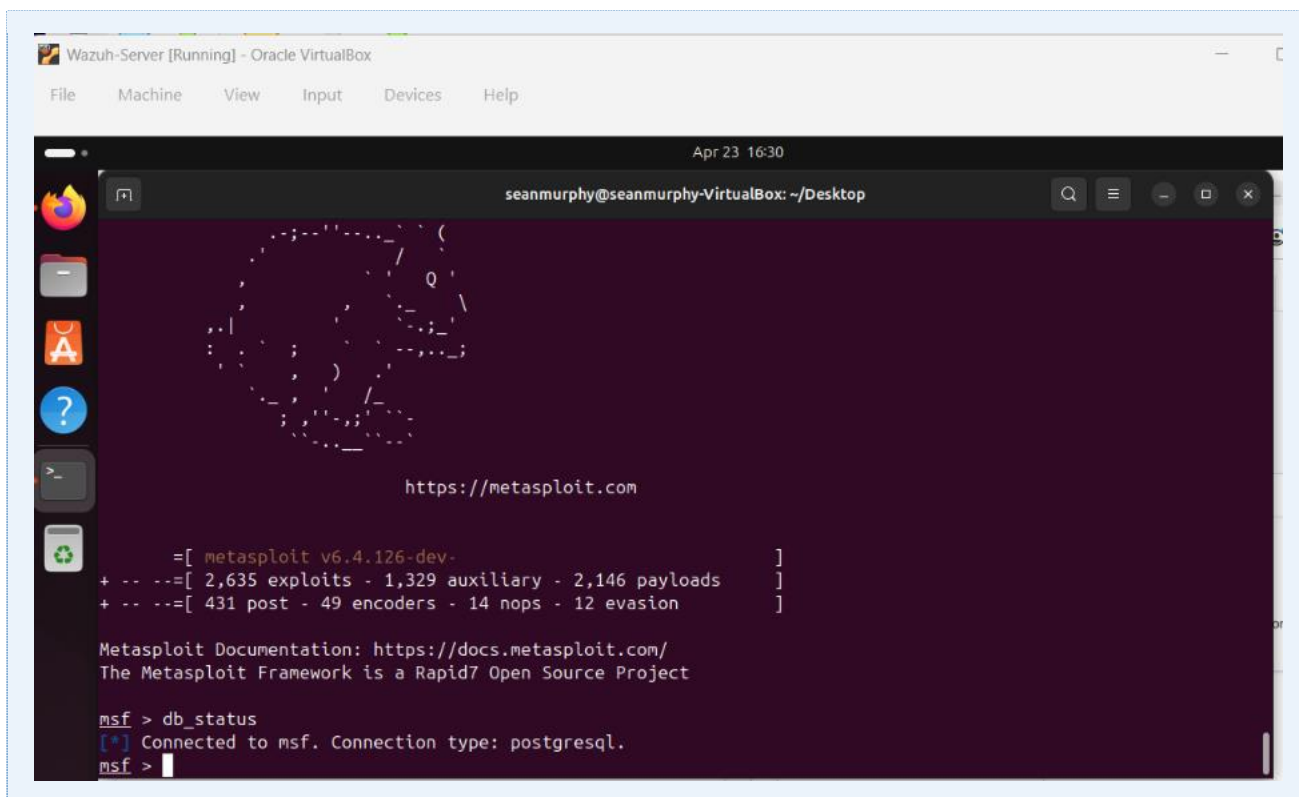


Figure: Metasploit Framework v6.4 console running on Ubuntu 24.04 with PostgreSQL database connected. 2,635 exploits and 1,329 auxiliary modules available. Attack simulation environment fully operational.

3. Firewall Configuration

3.1 Temporary Inbound Rule

Windows Defender Firewall was filtering all inbound traffic by default, blocking Metasploit from reaching SMB (port 445), RDP (port 3389), and WinRM (port 5985). A temporary inbound rule was created to allow traffic from the VirtualBox NAT subnet for the duration of the lab.

Security Note: This rule was scoped to specific ports and was deleted immediately after the lab concluded. The VirtualBox NAT subnet (10.0.2.0/24) is not reachable from the internet or from other devices on the home network. The exposure introduced was theoretical only.

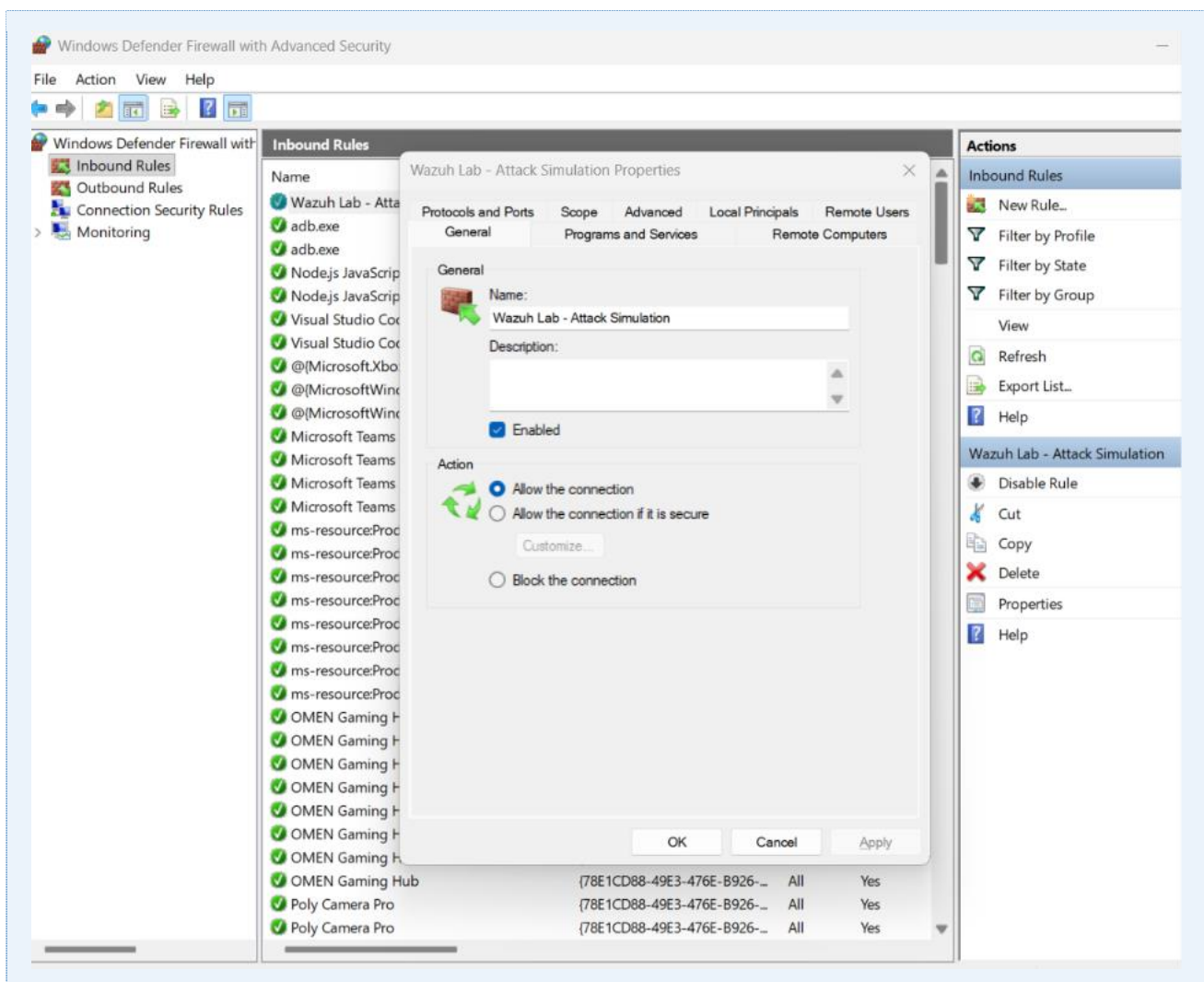


Figure: Temporary inbound firewall rule 'Wazuh Lab - Attack Simulation' created in Windows Defender Firewall. Rule allows inbound TCP on ports 445, 3389, and 5985. Rule was deleted upon lab completion, restoring the original firewall state.

3.2 NAT Gateway Discovery

An important networking discovery was made during setup: the Windows host was not directly reachable at its LAN IP (192.168.68.85) from the VirtualBox NAT network. Traffic had to be routed through the VirtualBox NAT gateway address (10.0.2.2), which maps directly to the Windows host. All subsequent attack scenarios used 10.0.2.2 as the target address.

```
# Port scan to identify reachable services
use auxiliary/scanner/portscan/tcp
set RHOSTS 10.0.2.2
set PORTS 445,3389,5985,139,135
set THREADS 4
run

[+] 10.0.2.2 - 10.0.2.2:135 - TCP OPEN
[+] 10.0.2.2 - 10.0.2.2:445 - TCP OPEN
```

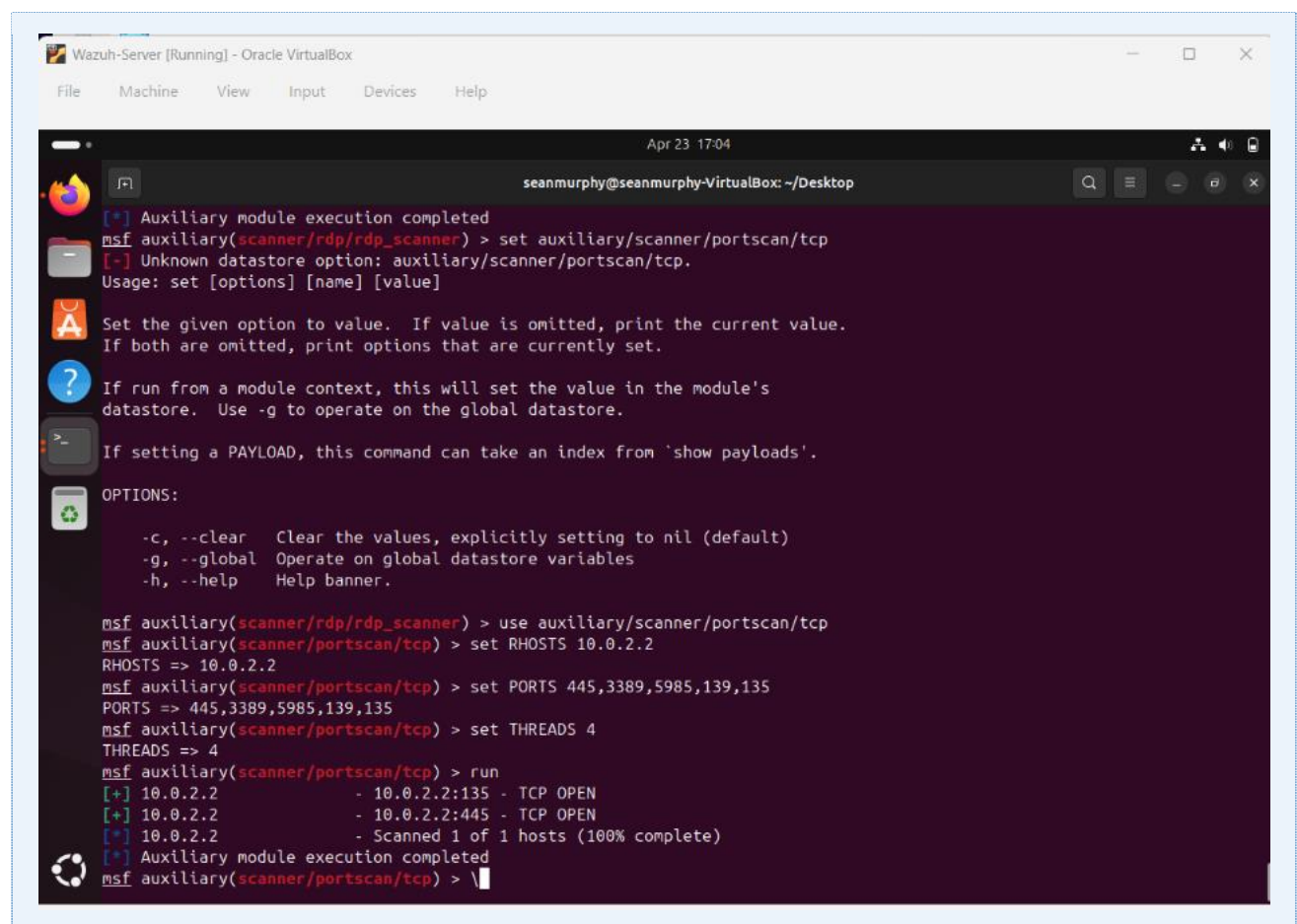
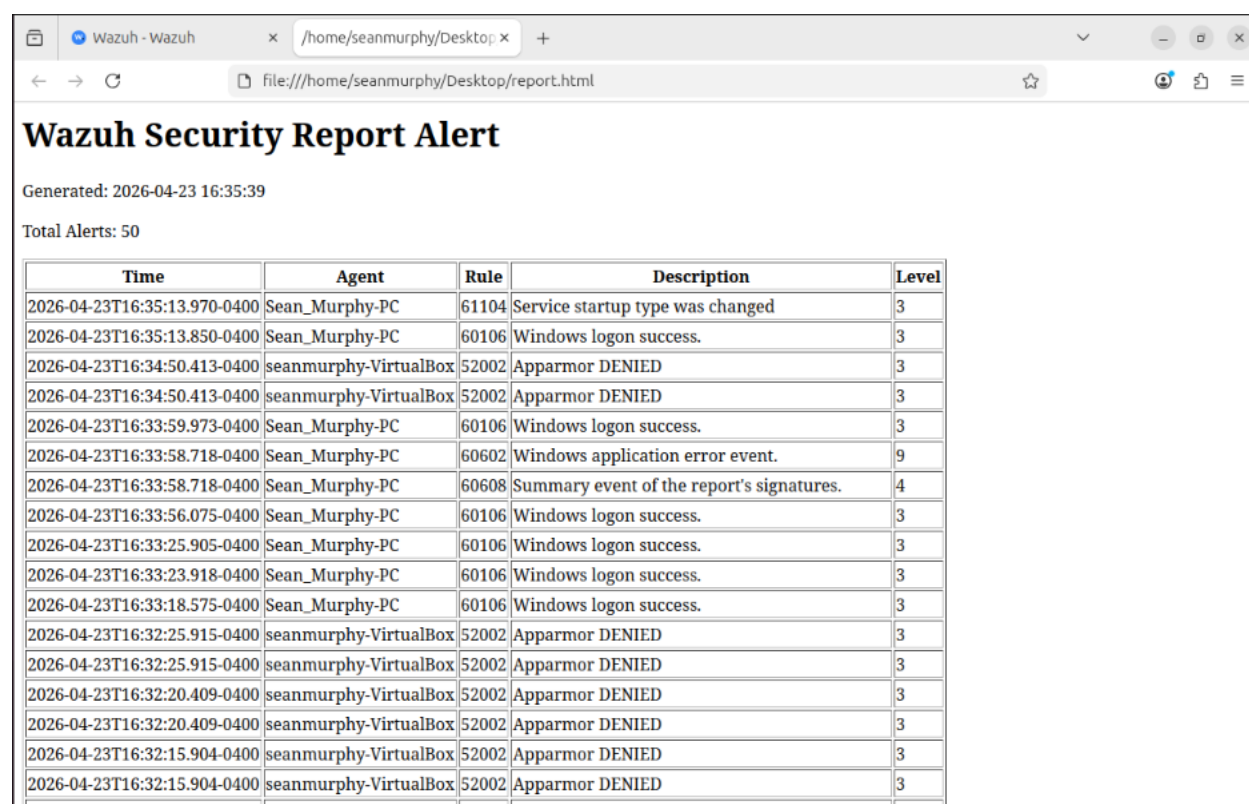


Figure: Metasploit TCP port scanner identifies open ports on Windows 11 host via VirtualBox NAT gateway (10.0.2.2). Ports 135 (RPC) and 445 (SMB) confirmed open — the attack surface for subsequent Metasploit scenarios.

4. Pre-Attack Baseline

Before executing any attack scenarios, the Python automated alert reporter was run to capture a baseline of existing Wazuh alerts. This establishes a before/after comparison that clearly demonstrates the impact of each attack scenario on the SIEM's alert volume and severity distribution.

The baseline report showed normal Windows operational events from Sean_Murphy-PC — Windows logon successes, service startup events, and AppArmor DENIED entries from the Ubuntu VM triggered by the Metasploit and Nmap installation processes. No authentication failure events were present.



Time	Agent	Rule	Description	Level
2026-04-23T16:35:13.970-0400	Sean_Murphy-PC	61104	Service startup type was changed	3
2026-04-23T16:35:13.850-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-23T16:34:50.413-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:34:50.413-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:33:59.973-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-23T16:33:58.718-0400	Sean_Murphy-PC	60602	Windows application error event.	9
2026-04-23T16:33:58.718-0400	Sean_Murphy-PC	60608	Summary event of the report's signatures.	4
2026-04-23T16:33:56.075-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-23T16:33:25.905-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-23T16:33:23.918-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-23T16:33:18.575-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-23T16:32:25.915-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:32:25.915-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:32:20.409-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:32:20.409-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:32:15.904-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:32:15.904-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3

Figure: Wazuh alert baseline captured via Python automated reporter before any attack simulation. Report shows normal Windows operational events from Sean_Murphy-PC — no authentication failures or brute force indicators present.

5. Attack Simulation Scenarios

Scenario 1 — Network Reconnaissance (Nmap)

Objective

Simulate the initial reconnaissance phase of an attack — an attacker mapping the network to identify live hosts, open ports, and running services before selecting an attack vector.

Command Executed

```
sudo nmap -sV -T4 192.168.68.85
```

Result

The service version scan confirmed the Windows host was alive with 0.0018s latency. All 1000 common TCP ports were in a filtered state, indicating Windows Firewall was actively dropping inbound packets. While no services were enumerated via this route, the scan confirmed host reachability and established the reconnaissance footprint.

Detection Value

Network reconnaissance attempts generate Windows Firewall audit events when firewall auditing is enabled. The scan traffic is also visible in Wazuh's network-level monitoring and contributes to the attack timeline documented in the dashboard.

Scenario 2 — Vulnerability Scanning (Nmap NSE Scripts)

Objective

Simulate a more aggressive reconnaissance phase using Nmap's built-in vulnerability detection scripts (NSE). This mirrors how a real attacker would probe for known CVEs and misconfigurations after initial host discovery.

Command Executed

```
sudo nmap --script vuln -T4 10.0.2.2
```

Result

The vulnerability scan identified three open ports and produced a significant finding on port 443:

- Port 135/tcp — Microsoft RPC

- Port 443/tcp — HTTPS (Wazuh dashboard)
- Port 445/tcp — Microsoft-DS (SMB)

On port 443, Nmap flagged an HTTP verb tampering vulnerability (CAPEC-274):

```
http-method-tamper:
  VULNERABLE:
    Authentication bypass by HTTP verb tampering
    State: VULNERABLE (Exploitable)
    This web server contains password protected resources vulnerable
    to authentication bypass vulnerabilities via HTTP verb tampering.

  URIs suspected to be vulnerable:
    /%5C1quot [POST]

  References:
    http://capec.mitre.org/data/definitions/274.html
    https://www.owasp.org/index.php/Testing_for_HTTP_Methods_and_XST
```

Analysis: This finding was detected on the Wazuh dashboard's HTTPS interface (port 443). The vulnerability indicates the web server may allow authentication bypass via non-standard HTTP methods such as PUT or DELETE. This is a real, actionable finding worth noting — though in a production environment, the Wazuh dashboard should be isolated to an internal-only network segment.

```
seanmurphy@seanmurphy-VirtualBox: ~/Desktop
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ sudo nmap --script vuln -T4 10.0.2.2
[sudo] password for seanmurphy:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-23 17:25 EDT
Nmap scan report for _gateway (10.0.2.2)
Host is up (0.0039s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
443/tcp    open  https
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
| http-method-tamper:
|   VULNERABLE:
|   Authentication bypass by HTTP verb tampering
|   State: VULNERABLE (Exploitable)
|   This web server contains password protected resources vulnerable to authentication bypass
|   vulnerabilities via HTTP verb tampering. This is often found in web servers that only limit access to the
|   common HTTP methods and in misconfigured .htaccess files.
|
|   Extra information:
|
|   URIs suspected to be vulnerable to HTTP verb tampering:
|   /%5C1quot [POST]
|
|   References:
|   http://capec.mitre.org/data/definitions/274.html
|   https://www.owasp.org/index.php/Testing_for_HTTP_Methods_and_XST_%28OWASP-CM-008%29
|   http://www.imperva.com/resources/glossary/http_verb_tampering.html
|   http://www.mkit.com.ar/labs/htexploit/
|_
445/tcp    open  microsoft-ds
MAC Address: 52:55:0A:00:02:02 (Unknown)
```

Figure: Nmap vulnerability script scan (--script vuln) against Windows 11 host via NAT gateway. Scanner identifies open ports 135 (RPC), 443 (HTTPS), and 445 (SMB). Real finding: HTTP verb tampering vulnerability (CAPEC-274, OWASP reference) detected on port 443 — the Wazuh dashboard interface.

Scenario 3 — SMB Credential Brute Force (Metasploit)

Objective

Simulate a credential brute force attack against the Windows SMB service. This is one of the most common real-world attack patterns and the scenario most likely to generate high-severity Wazuh alerts. SMB brute forcing generates Windows Security Event ID 4625 (failed logon) for each attempt, which Wazuh's rule engine correlates into brute force alerts.

MITRE ATT&CK Mapping

ATT&CK Field	Value
Tactic	Credential Access
Technique	T1110 — Brute Force
Sub-technique	T1110.003 — Password Spraying
Target	SMB (port 445) — Administrator account

Commands Executed

```
use auxiliary/scanner/smb/smb_login
set RHOSTS 10.0.2.2
set SMBUser Administrator
set PASS_FILE /opt/metasploit-
framework/embedded/framework/data/wordlists/unix_passwords.txt
set THREADS 4
run
```

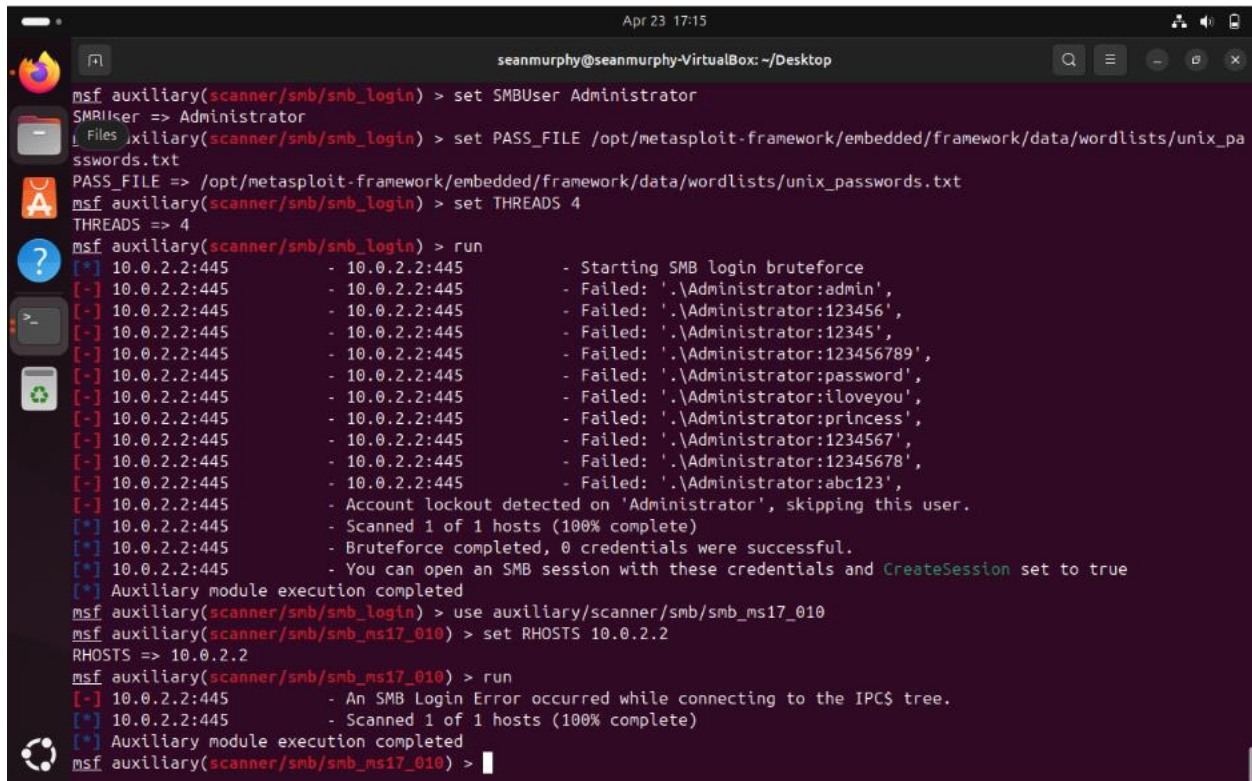
Result

Metasploit successfully connected to SMB on port 445 and began firing authentication attempts from the unix_passwords.txt wordlist. The following sequence of events occurred:

- Starting SMB login bruteforce against 10.0.2.2:445
- Failed: .\Administrator:admin
- Failed: .\Administrator:123456
- Failed: .\Administrator:12345
- Failed: .\Administrator:123456789
- Failed: .\Administrator:password
- Failed: .\Administrator:iloveyou
- Failed: .\Administrator:princess
- Failed: .\Administrator:1234567
- Failed: .\Administrator:12345678
- Failed: .\Administrator:abc123

- Account lockout detected on 'Administrator' — skipping this user

Key Outcome: Windows locked the Administrator account after detecting repeated failed authentication attempts. This triggered Windows Security Event ID 4740 (account locked out) in addition to the stream of 4625 events. Wazuh detected and categorized all of these events in real time, firing its brute force correlation rules.



```

msf auxiliary(scanner/smb/smb_login) > set SMBUser Administrator
SMBUser => Administrator
msf auxiliary(scanner/smb/smb_login) > set PASS_FILE /opt/metasploit-framework/embedded/framework/data/wordlists/unix_passwords.txt
PASS_FILE => /opt/metasploit-framework/embedded/framework/data/wordlists/unix_passwords.txt
msf auxiliary(scanner/smb/smb_login) > set THREADS 4
THREADS => 4
msf auxiliary(scanner/smb/smb_login) > run
[*] 10.0.2.2:445 - 10.0.2.2:445 - Starting SMB login brute force
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:admin',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:123456',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:12345',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:123456789',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:password',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:iloveyou',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:princess',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:1234567',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:12345678',
[*] 10.0.2.2:445 - 10.0.2.2:445 - Failed: '.\Administrator:abc123',
[*] 10.0.2.2:445 - Account lockout detected on 'Administrator', skipping this user.
[*] 10.0.2.2:445 - Scanned 1 of 1 hosts (100% complete)
[*] 10.0.2.2:445 - Brute force completed, 0 credentials were successful.
[*] 10.0.2.2:445 - You can open an SMB session with these credentials and CreateSession set to true
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_login) > use auxiliary/scanner/smb/smb_ms17_010
msf auxiliary(scanner/smb/smb_ms17_010) > set RHOSTS 10.0.2.2
RHOSTS => 10.0.2.2
msf auxiliary(scanner/smb/smb_ms17_010) > run
[*] 10.0.2.2:445 - An SMB Login Error occurred while connecting to the IPC$ tree.
[*] 10.0.2.2:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_ms17_010) >

```

Figure: Metasploit smb_login auxiliary module executing SMB brute force against Windows 11 host. Multiple failed authentication attempts against the Administrator account trigger Windows Security Event ID 4625 (failed logon). Attack culminates in Windows Security Event ID 4740 — account lockout — detected and reported by Wazuh.

Attack Chain Interaction

An important real-world phenomenon was observed: the Administrator account lockout caused by the brute force then blocked a subsequent EternalBlue vulnerability scan (MS17-010), which requires SMB authentication to probe the IPC\$ tree. This demonstrates how attack sequences interact with defensive controls — a realistic behavior that SOC analysts encounter when correlating multi-stage incidents.

```

use auxiliary/scanner/smb/smb_ms17_010
set RHOSTS 10.0.2.2
run

```

```
[~] 10.0.2.2:445 - An SMB Login Error occurred while connecting to the  
IPC$ tree.  
[*] 10.0.2.2:445 - Scanned 1 of 1 hosts (100% complete)
```

Portfolio Note: This interaction — brute force causing lockout causing a subsequent scan to fail — is more valuable than a clean result. It demonstrates understanding of how attack stages affect each other and how a defender needs to think about the full attack timeline, not just individual events.

6. Wazuh Detection Results

6.1 Security Events Dashboard

Following the attack simulation scenarios, the Wazuh Security Events dashboard was reviewed. The dashboard showed a dramatic spike in alert activity corresponding precisely to the SMB brute force window:

Metric	Value
Total Alerts (24hr period)	277
Level 12 or Above Alerts	0
Authentication Failures	11
Authentication Successes	22
Top MITRE Tactic Detected	Brute Force
Other MITRE Categories	Valid Accounts, Account Access Restrictions, Sudo and Sudo Caching

The alert level evolution chart showed a sharp spike at the 15:00-16:00 window corresponding directly to the Metasploit brute force execution time. The MITRE ATT&CK donut chart automatically categorized the attack activity.

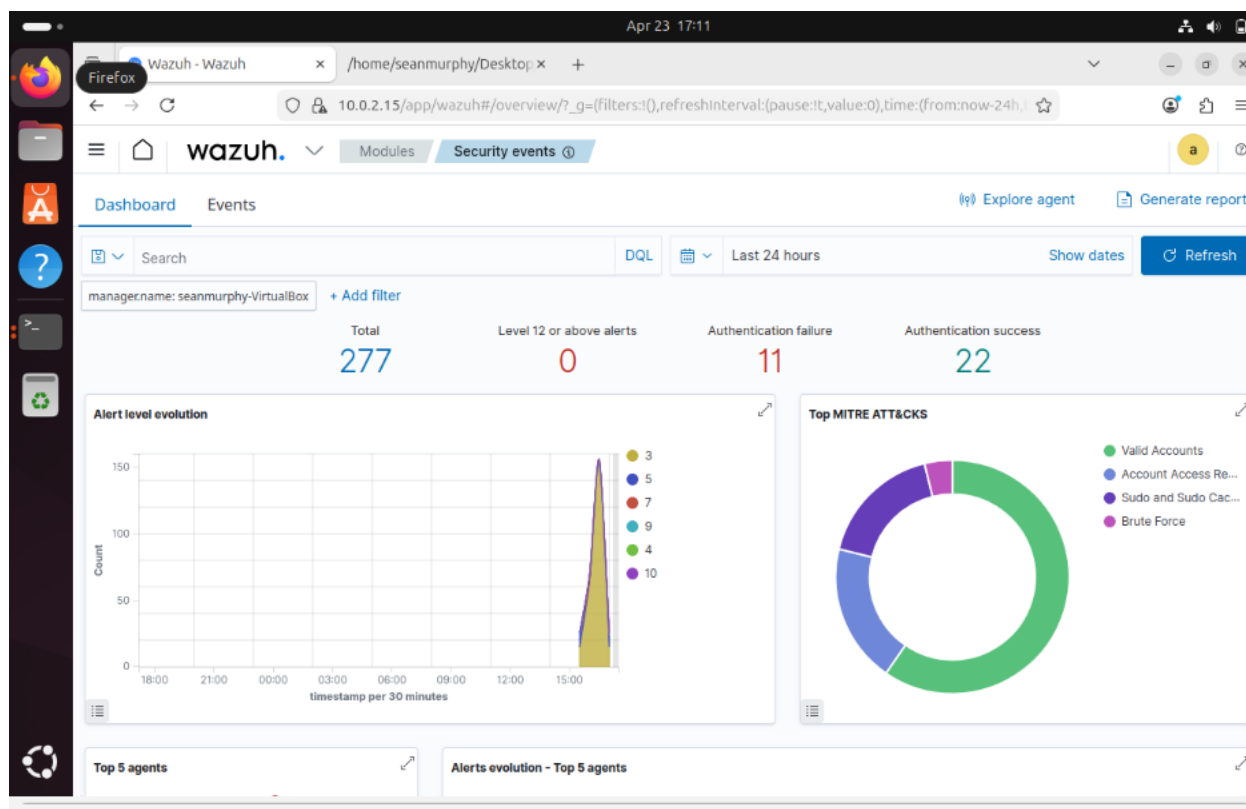


Figure: Wazuh Security Events dashboard following attack simulation. 277 total alerts generated with a visible spike corresponding to the SMB brute force window. MITRE ATT&CK wheel automatically categorizes detected activity under Brute Force, Account Access Restrictions, and Valid Accounts tactics.

6.2 Key Alert Rules Triggered

Rule ID	Description & Significance
60122	Logon failure — Unknown user or bad password. Severity Level 5. Fired for every individual failed SMB authentication attempt from Metasploit's wordlist.
60204	Multiple Windows logon failures. Severity Level 10. Wazuh's correlation rule — fires when repeated 4625 events from the same source exceed a threshold within a time window. The core brute force detection.
60115	User account locked out (multiple login errors). Severity Level 9. Fired when Windows generated Event ID 4740 after the Administrator account was locked. Highest severity event of the simulation.
60106	Windows logon success. Level 3. Background noise — normal user sessions on the Windows host.
52002	AppArmor DENIED. Level 3. Wazuh VM's AppArmor catching tool invocations — Metasploit and Nmap triggering Linux security policies on the Ubuntu host.

6.3 Python Automated Alert Reporter

The Python script built in Part 1 was run before and after the attack simulation to capture the full impact. The post-attack report clearly shows the brute force event timeline, with rule 60122 (logon failure) entries firing in rapid succession followed by rule 60204 (multiple failures) and rule 60115 (account lockout).

The report is generated by authenticating to the Wazuh REST API on port 55000 via JWT token, then querying the OpenSearch indexer on port 9200 for the 50 most recent alerts from the wazuh-alerts-* index. Alert fields are extracted from the JSON response and written to a formatted HTML table.

Wazuh Security Report Alert

Generated: 2026-04-23 17:08:53

Total Alerts: 50

Time	Agent	Rule	Description	Level
2026-04-23T17:08:03.917-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:08:03.911-0400	Sean_Murphy-PC	60115	User account locked out (multiple login errors).	9
2026-04-23T17:08:01.674-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:07:59.447-0400	Sean_Murphy-PC	60204	Multiple Windows logon failures.	10
2026-04-23T17:07:57.175-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:07:54.995-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:07:52.784-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:07:50.564-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:07:48.222-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:07:46.195-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:07:43.710-0400	Sean_Murphy-PC	60122	Logon failure - Unknown user or bad password.	5
2026-04-23T17:03:24.627-0400	Sean_Murphy-PC	60642	Software protection service scheduled successfully.	3
2026-04-23T16:51:02.340-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:51:02.340-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-23T16:45:10.474-0400	Sean_Murphy-PC	60642	Software protection service scheduled successfully.	3
2026-04-23T16:44:45.787-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-23T16:44:25.044-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3

Figure: Python automated alert reporter (github.com/murphys611/wazuh-alert-reporter) captures Wazuh brute force detections in real time. Report shows rapid-fire rule 60122 (logon failure) events from Sean_Murphy-PC, escalating to rule 60204 (multiple failures, level 10) and rule 60115 (account lockout, level 9).

7. Summary & Key Takeaways

7.1 Scenarios Completed

Scenario	Outcome
Nmap Service Scan	Host confirmed alive. All ports filtered by default Windows Firewall — demonstrates firewall effectiveness.
Metasploit Port Scan	Ports 135 and 445 open via NAT gateway. Attack surface identified.
Nmap Vulnerability Scan	Real VULNERABLE finding on port 443 — HTTP verb tampering (CAPEC-274). Referenced by OWASP and MITRE.
SMB Brute Force	Metasploit successfully fired credential attacks. Windows logged 4625 events. Wazuh rules 60122, 60204, 60115 all fired. Account lockout triggered.
Attack Chain Interaction	Brute force lockout blocked subsequent EternalBlue scan — demonstrating realistic multi-stage attack behavior.

7.2 Key Learnings

- Wazuh's rule correlation engine effectively escalates individual failed logon events (level 5) into brute force alerts (level 10) when the frequency threshold is exceeded — this is the core value of a SIEM over raw log analysis.
- The MITRE ATT&CK integration in Wazuh automatically maps detected events to standardized tactics and techniques, providing immediate context for SOC analysts without manual correlation.
- VirtualBox NAT network routing requires using the gateway address (10.0.2.2) rather than the host's LAN IP for inbound connections — a networking constraint that affects real-world lab design.
- Attack sequences interact: the brute force lockout blocking the subsequent EternalBlue scan is a realistic example of how early attack stages affect later ones — important for incident timeline reconstruction.
- A real vulnerability (HTTP verb tampering) was discovered on the Wazuh dashboard itself during the Nmap vulnerability scan, demonstrating that even security tools can have misconfigurations worth investigating.

7.3 Skills Demonstrated

- Metasploit Framework — auxiliary module selection, option configuration, wordlist-based credential attacks
- Nmap — service version detection (-sV), NSE vulnerability scripts (--script vuln), timing templates (-T4)
- Wazuh SIEM — alert rule interpretation, MITRE ATT&CK dashboard, security events filtering

- Windows Security Events — Event ID 4625 (failed logon), 4740 (account lockout), 4624 (successful logon)
- Network fundamentals — NAT routing, port scanning, SMB protocol, firewall rule management
- Python automation — REST API authentication, OpenSearch queries, HTML report generation

7.4 Cleanup

All temporary changes made to the Windows host for this lab were reversed upon completion:

- Firewall rule 'Wazuh Lab - Attack Simulation' deleted from Windows Defender Firewall
- Administrator account lockout resolved
- No persistent changes remain on the Windows host

8. References & Resources

- Wazuh Documentation: <https://documentation.wazuh.com>
- Metasploit Framework: <https://docs.metasploit.com>
- Nmap NSE Scripts: <https://nmap.org/nsedoc/>
- MITRE ATT&CK T1110 (Brute Force): <https://attack.mitre.org/techniques/T1110/>
- CAPEC-274 (HTTP Verb Tampering): <https://capec.mitre.org/data/definitions/274.html>
- OWASP HTTP Methods Testing: <https://owasp.org/www-project-web-security-testing-guide/>
- Python Alert Reporter (GitHub): <https://github.com/murphys611/wazuh-alert-reporter>
- Windows Security Event IDs: <https://docs.microsoft.com/en-us/windows/security/threat-protection/auditing/>